

Hung-Ting Su

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Robot Learning • Embodied AI • VLA / World Models • Real-World Manipulation

Summary

- Researcher-engineer building reliable embodied AI systems for real-world robot manipulation and navigation, spanning multimodal sensing, VLA-style policies, rollout evaluation, and failure-driven improvement.
- Current work focuses on tactile/visual policy learning, robot-learning infrastructure, contact-rich manipulation, and emerging world-model directions for real-world decision-making.
- Prior work connects embodied navigation, mobile manipulation, policy monitoring, affordance-aware planning, and 3D robotic perception with trustworthy multimodal reasoning.

Experience

Senior Associate Researcher

2025–Present

Delta Robotics Innovation Center, Delta Electronics

Taiwan

- Build end-to-end robot-learning pipelines for manipulation, from multimodal data collection and curation to VLA-style policy training, rollout evaluation, and on-robot validation.
- Investigate tactile and visual policy learning for contact-rich manipulation, including failure analysis for slip, misalignment, under-force, and observation mismatch.
- Develop embodied-AI prototypes that connect language, perception, planning, action, and emerging world-model-oriented reasoning for robust real-world execution.

Postdoctoral Researcher

2023–2025

National Taiwan University (NTU)

Taiwan

- Led and mentored projects on embodied navigation, mobile manipulation, policy monitoring, false-premise navigation, and affordance-aware planning.
- Built evaluation and dataset pipelines for embodied AI, including benchmark design, failure-mode analysis, and zero-shot VLM-based replanning workflows.
- Led projects resulting in CoRL 2024, EMNLP 2024, and ACL 2026, and mentored student-led robotics work accepted at NeurIPS 2024, AAAI 2026, and ACL 2026.

Visiting Researcher

2022–2023

Columbia University, hosted by Prof. Shih-Fu Chang

New York, NY

- Developed early zero-shot causal video reasoning frameworks and maintained a strong bridge between embodied AI and trustworthy multimodal reasoning.
- Initiated and coordinated a Columbia–NTU collaboration that led to an EMNLP 2024 publication.

Selected Robot Learning Publications

Full publication history: [Google Scholar](#).

- **VLN-NF: Feasibility-Aware Vision-and-Language Navigation with False-Premise Instructions** (*ACL 2026*) [paper] [project page] — Evidence-seeking embodied navigation with explicit abstention under false-premise or infeasible instructions. **Role: Lead**
- **Context-Aware Replanning with Pre-Explored Semantic Map for Object Navigation** (*CoRL 2024*) [paper] [code] — Zero-shot visual map refinement and semantic replanning for more robust embodied navigation. **Role: Lead**
- **AED: Adaptable Error Detection for Few-shot Imitation Policy** (*NeurIPS 2024*) [paper] — Policy monitoring for few-shot imitation learning with deployment-oriented error detection. **Role: Mentor**
- **Affordance-Guided Coarse-to-Fine Exploration for Base Placement in Open-Vocabulary Mobile Manipulation** (*AAAI 2026*) [paper] — Training-free VLM-guided planning for mobile-manipulation base placement in open-vocabulary settings. **Role: Mentor**
- **ADAPT: Benchmarking Commonsense Planning under Unspecified Affordance Constraints** (*ACL 2026*) [paper] — Benchmarking and evaluating planning under under-specified affordance and feasibility constraints. **Role: Mentor**
- **Unveiling Narrative Reasoning Limits of Large Language Models with Trope in Movie Synopses** (*EMNLP 2024*) [paper] [code] — Analyzes narrative reasoning failure modes in LLMs and reduces unsupported chain-of-thought behavior. **Role: Lead**
- **Stage Conscious Attention Network (SCAN) : A Demonstration-Conditioned Policy for Few-Shot Imitation** (*AAAI 2022*) [paper] — Few-shot imitation learning through stage-aware policy conditioning from limited demonstrations. **Role: Mentor**
- **A Coarse-To-Fine (C2F) Representation for End-To-End 6-DoF Grasp Detection** (*CoRL 2020*) [paper] — 6-DoF grasp detection in cluttered scenes with a coarse-to-fine representation for manipulation.

Education

PhD in Computer Science

2023

National Taiwan University (NTU), Advisor: Prof. Winston H. Hsu

Taiwan

- Dissertation: *Multi-modal Video Comprehension and Beyond*
- Visiting researcher at Columbia University, hosted by Prof. Shih-Fu Chang.

MS in Computer Science

2017

National Taiwan University

Taiwan

BS in Computer Science

2013

National Tsing Hua University

Taiwan

Skills

- **Robot Learning / Embodied AI:** imitation learning, diffusion policies, VLA-style policies, world-model-oriented planning, policy monitoring, failure analysis, and deployment-oriented evaluation.
- **Manipulation / Navigation / Planning:** mobile manipulation, base placement, embodied navigation, affordance-aware planning, evidence-seeking exploration, and false-premise instruction handling.
- **Multimodal Sensing:** RGB-D, 3D point clouds, tactile/contact feedback, open-vocabulary perception, and sensor-driven failure analysis.

- **Systems / Robotics Stack:** Python, PyTorch, ROS2, Linux, Docker, Git, Isaac Sim, and real-robot rollout/data pipelines.

Academic Services

- Area Chair: *EACL 2026, ACL 2026*
- Reviewer: *ICML, NeurIPS, CVPR, ICCV, NAACL, ACL, EMNLP, CoRL, ICRA, IROS, AAAI, ACM Multimedia, TCSVT, and related venues*

References

References are available upon request.